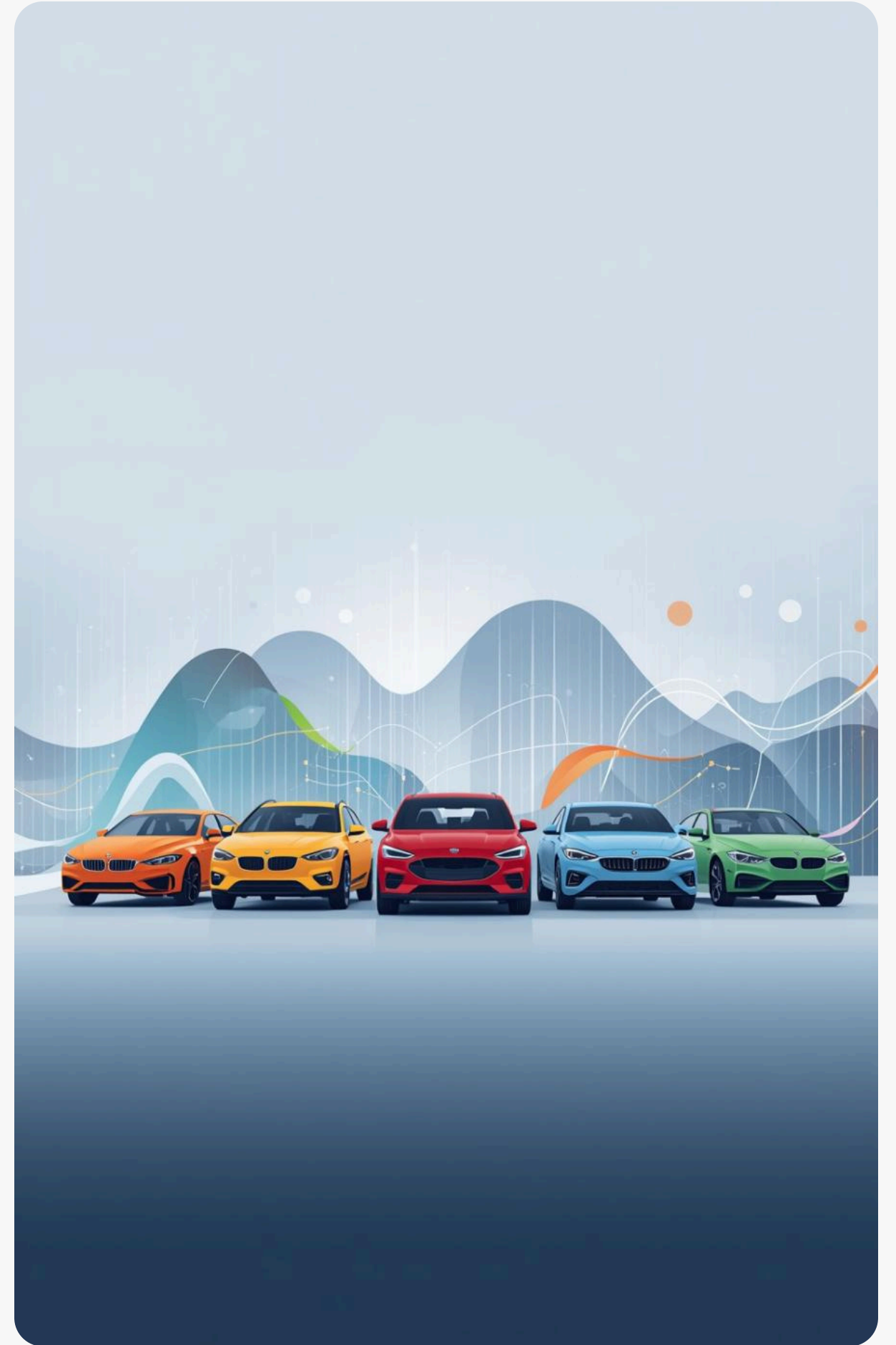


Car Price Prediction

A machine learning portfolio project

Presented by:

Annelize Krause



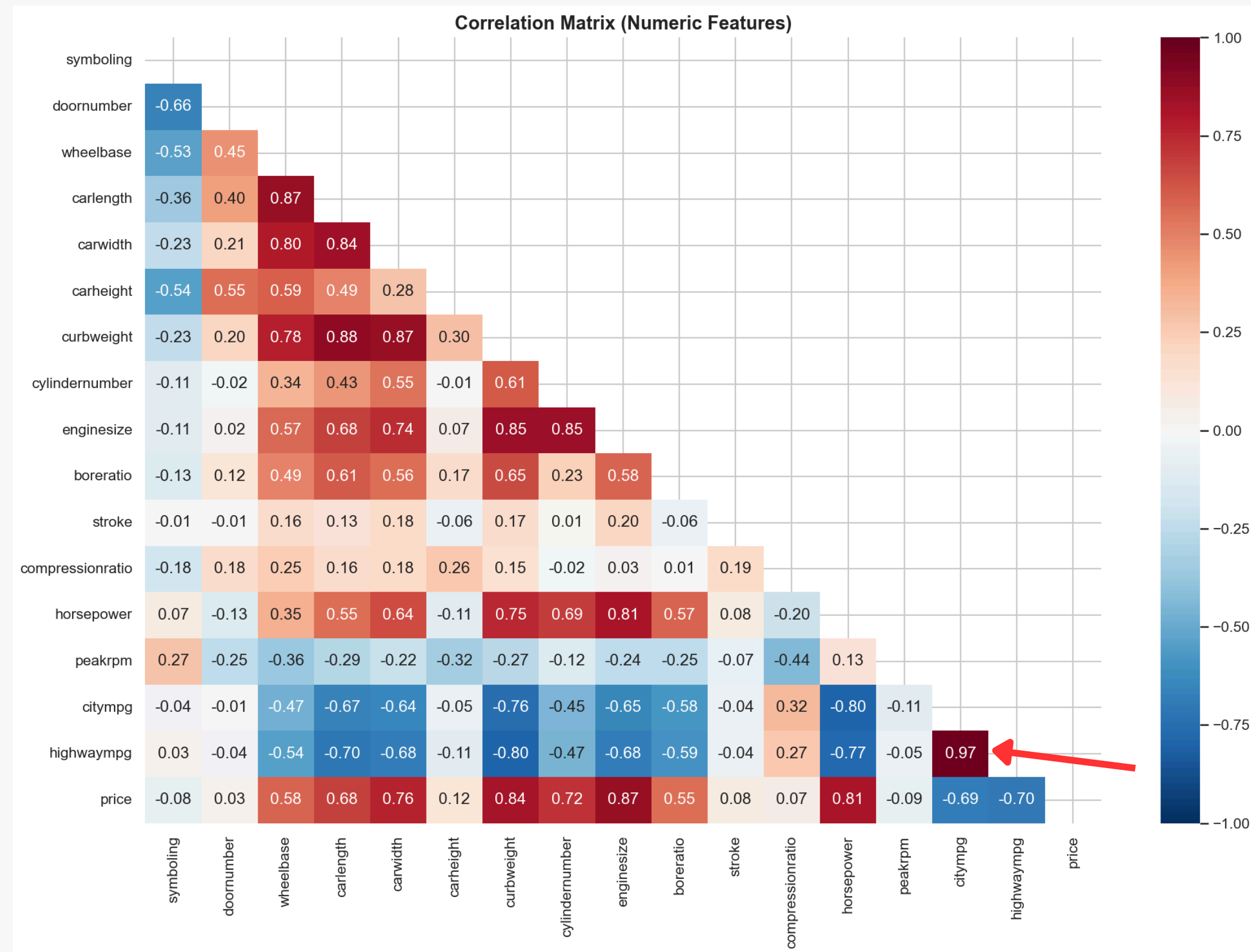
Data Preparation & Adding Features

Kaggle Dataset:

- 205 cars
- 24 features (engine size, weight, fuel type, etc.)

Feature Engineering:

- Merged near-identical city and highway fuel economy into **avg_mpg**
- Combined horsepower and weight into a power-to-weight ratio (**hp_per_weight**)
- Extracted 22 brands from the car name field
- Grouped brands into **brand_tier** by median price:
 - budget < \$8,916
 - mid < \$16,503
 - premium >= \$16,503



Manual Models

Selection:

- Linear Regression: Straight-line baseline
- Random Forest: Ensemble of trees, captures non-linear patterns
- Gradient Boosting: Sequential trees, each correcting the last

Metrics:

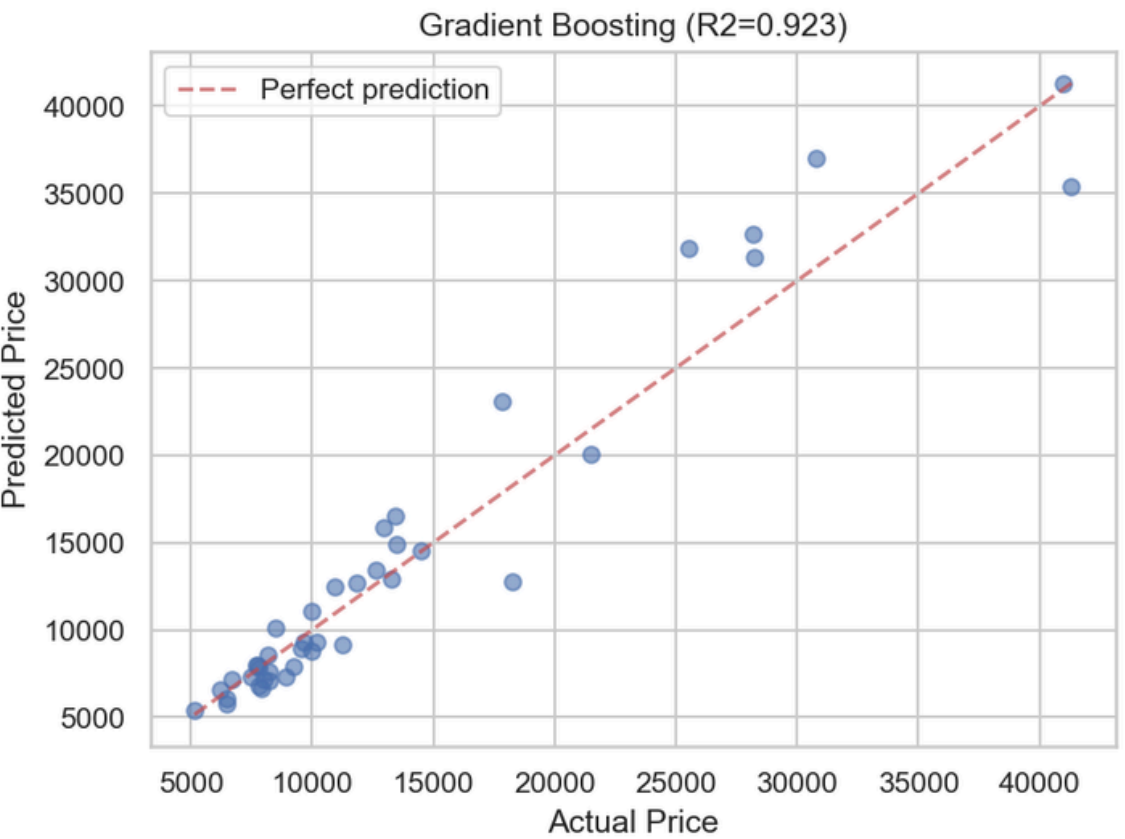
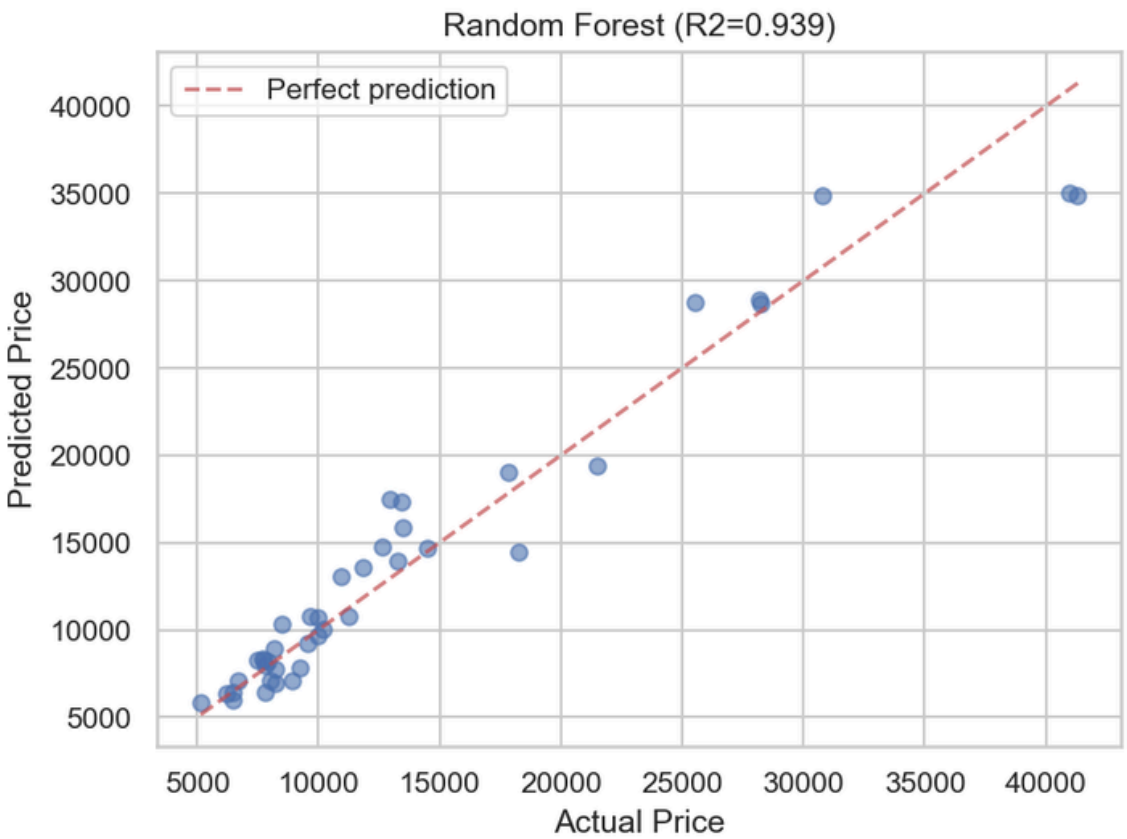
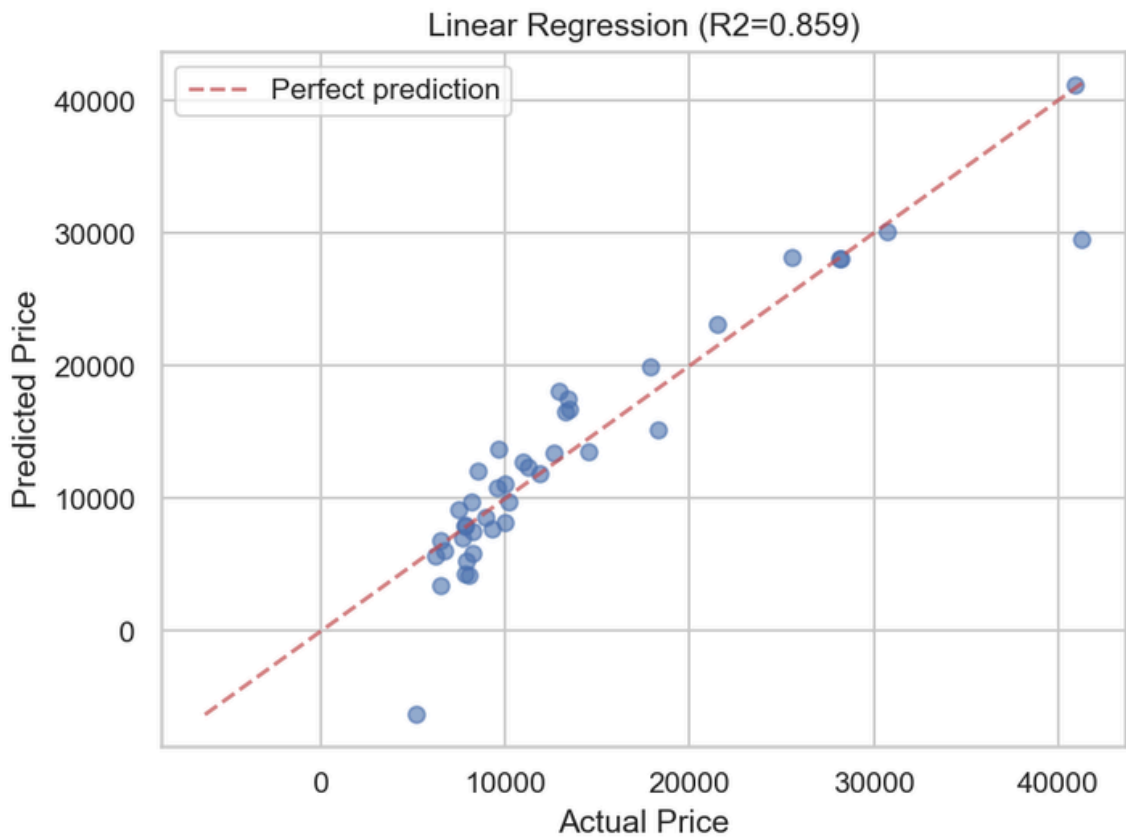
- Mean Absolute Error (MAE): Average amount the model is off by
- Root Mean Square Error (RMSE): Penalises big misses more heavily
- R-squared (R^2): Proportion of price variation explained



Manual Model Evaluation

Model	MAE	RMSE	R ²
Random Forest	\$1,557	\$2,208	0.938
Gradient Boosting	\$1,692	\$2,545	0.918
Linear Regression	\$2,264	\$3,329	0.860

Actual vs Predicted Price



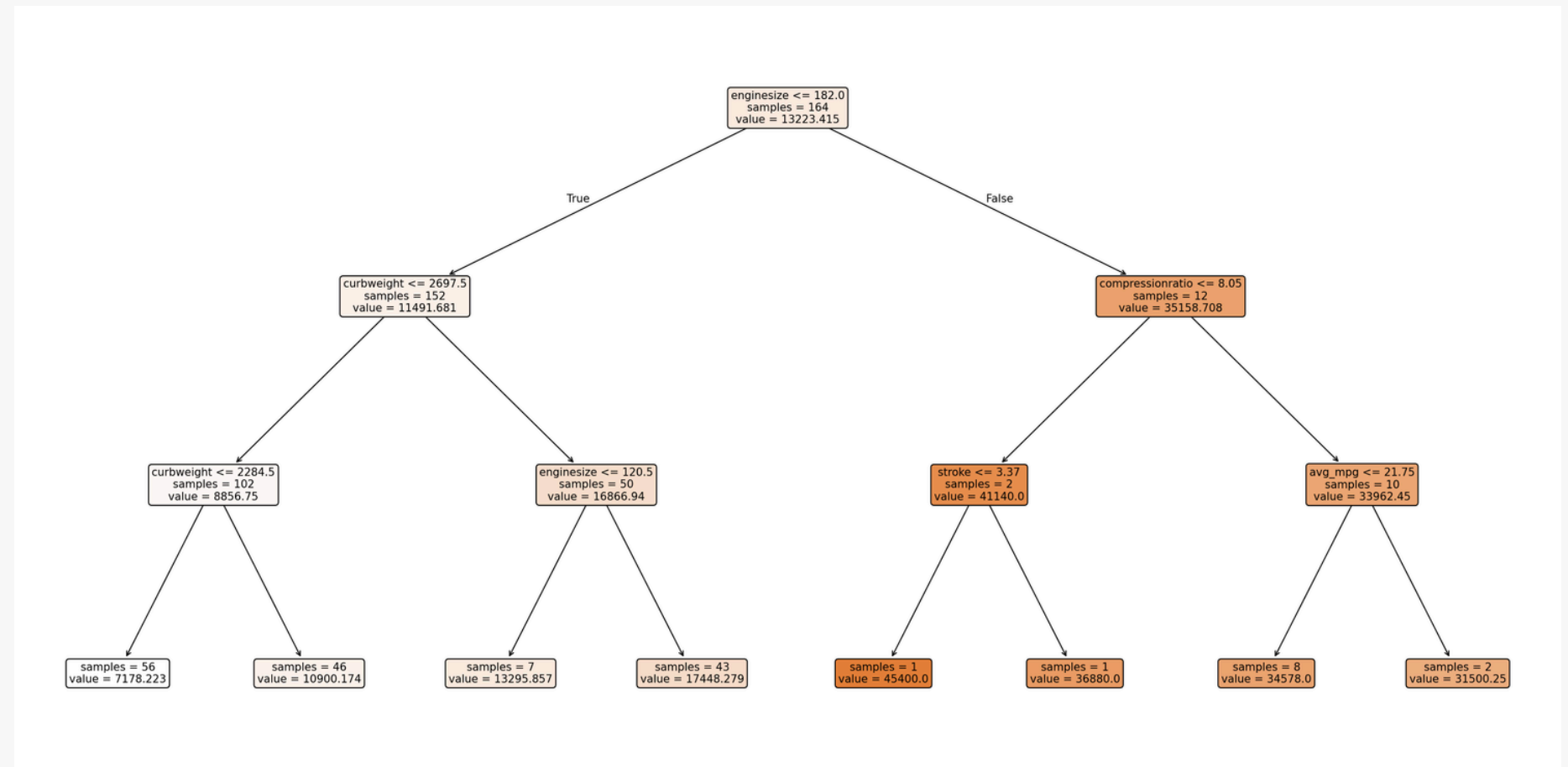
White-Box Interpretation

Linear Regression:

- **curbweight**: +\$6.36 per pound
- **fueltype**: +\$12,040 (diesel vs gas)
- Spotted the idi **fuelsystem** was encoding the same info as diesel **fueltype** - dropped **fuelsystem** entirely

Decision Tree (depth 3):

- Top features:
 - enginesize (70.6%)
 - curbweight (27.9%)
- Root split: enginesize ≤ 182



PyCaret Model - Raw Data

PyCaret Pick (cross-validation):
Light Gradient Boosting Machine

Best Model on Test Set:
Random Forest

Model	MAE	RMSE	R ²
Random Forest Regressor	\$1,522	\$2,128	0.943
Extra Trees Regressor	\$1,628	\$2,334	0.931
Gradient Boosting Regressor	\$1,811	\$2,609	0.914
Decision Tree Regressor	\$1,817	\$2,549	0.918
AdaBoost Regressor	\$1,986	\$2,522	0.919
Light Gradient Boosting Machine	\$2,177	\$3,558	0.840
Ridge Regression	\$2,286	\$3,582	0.837
Linear Regression	\$2,410	\$3,752	0.822
Lasso Least Angle Regression	\$2,467	\$3,693	0.827
Lasso Regression	\$2,467	\$3,693	0.827
Elastic Net	\$2,500	\$3,568	0.839
Bayesian Ridge	\$2,722	\$3,781	0.819
Orthogonal Matching Pursuit	\$2,740	\$3,787	0.818
Huber Regressor	\$2,836	\$4,486	0.745
K Neighbors Regressor	\$2,868	\$4,916	0.694

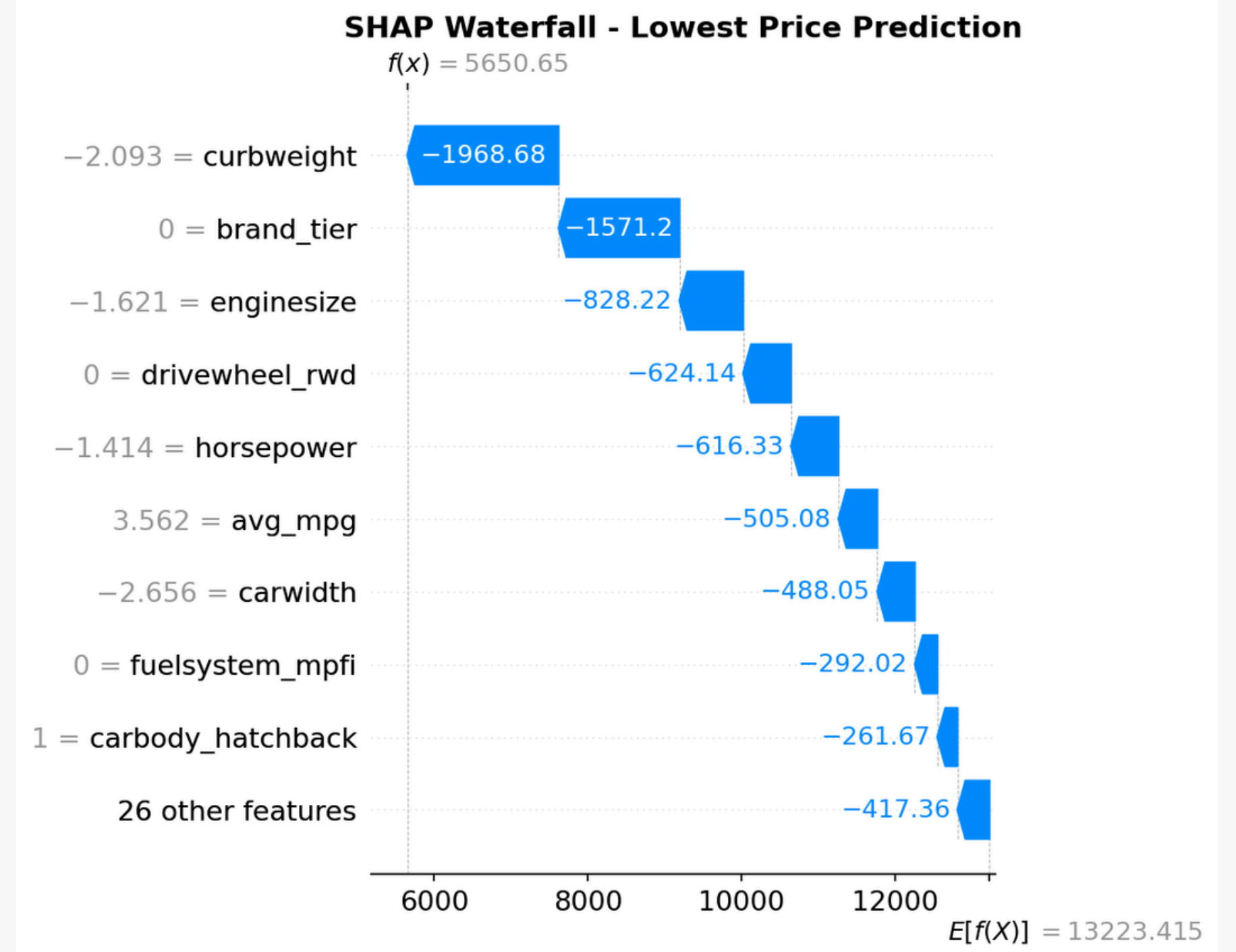
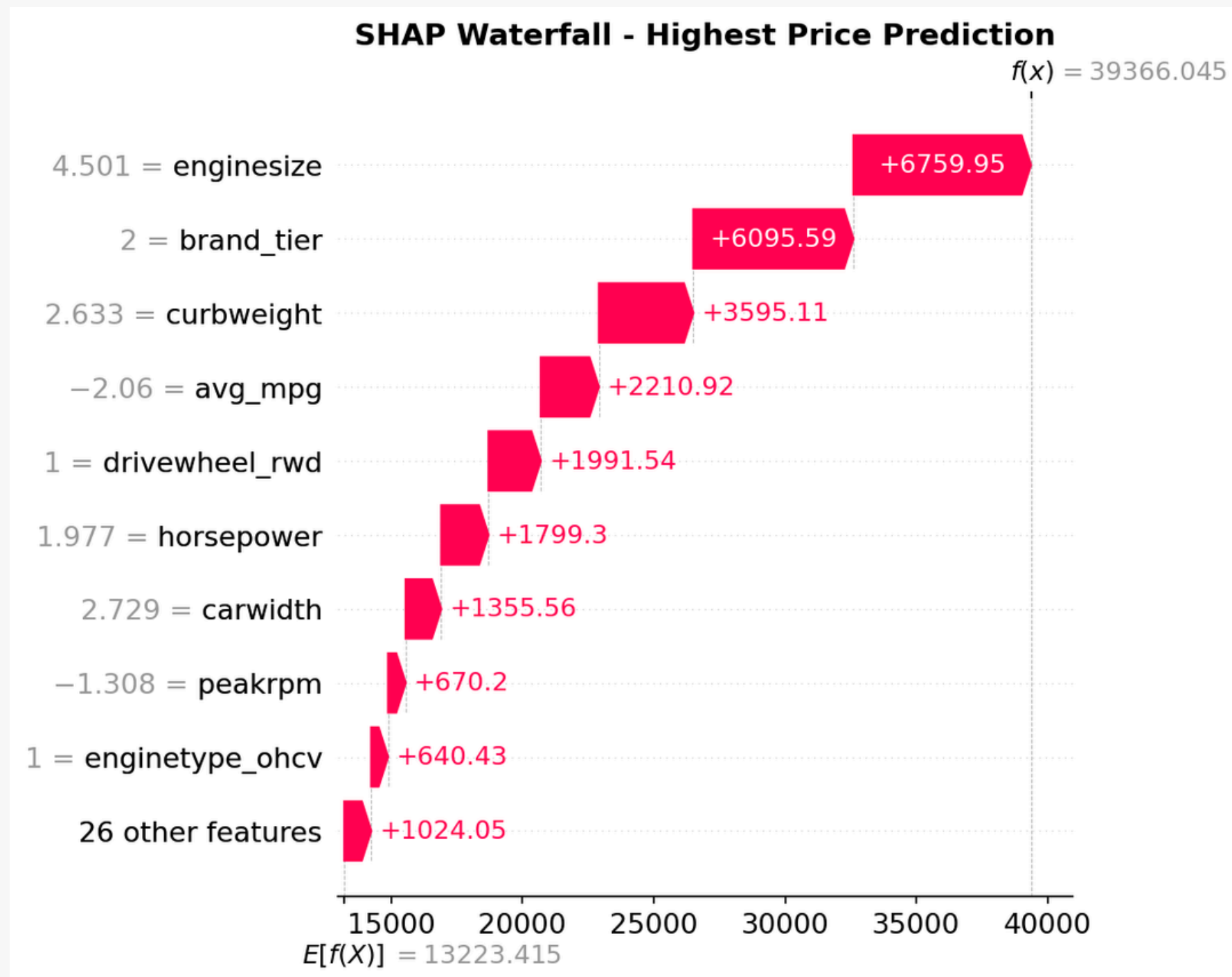
PyCaret Model - Preprocessed Data

PyCaret Pick (cross-validation):
Extra Trees

Best Model on Test Set:
Random Forest

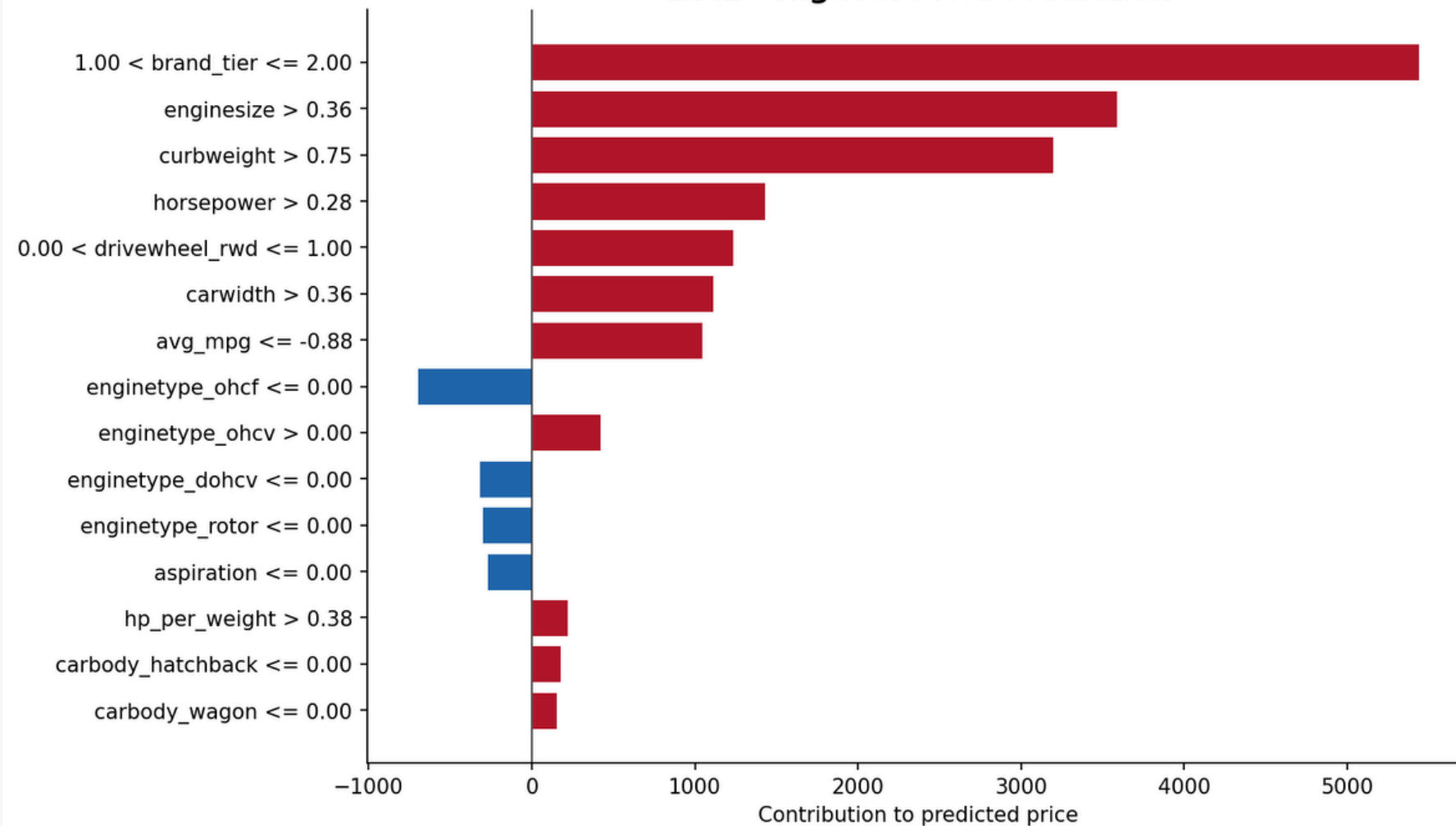
Model	MAE	RMSE	R²
Random Forest Regressor	\$1,280	\$1,820	0.958
Extra Trees Regressor	\$1,541	\$2,288	0.934
Gradient Boosting Regressor	\$1,654	\$2,285	0.934
Decision Tree Regressor	\$1,849	\$2,570	0.916
AdaBoost Regressor	\$1,899	\$2,529	0.919
Huber Regressor	\$2,029	\$3,076	0.880
Light Gradient Boosting Machine	\$2,087	\$3,264	0.865
Ridge Regression	\$2,228	\$3,354	0.858
Bayesian Ridge	\$2,231	\$3,360	0.857
Lasso Least Angle Regression	\$2,243	\$3,313	0.861
Lasso Regression	\$2,243	\$3,313	0.861
Linear Regression	\$2,264	\$3,329	0.860
Orthogonal Matching Pursuit	\$2,509	\$3,448	0.849
Passive Aggressive Regressor	\$2,574	\$3,590	0.837
Elastic Net	\$2,642	\$3,930	0.804

Understanding Predictions with SHAP

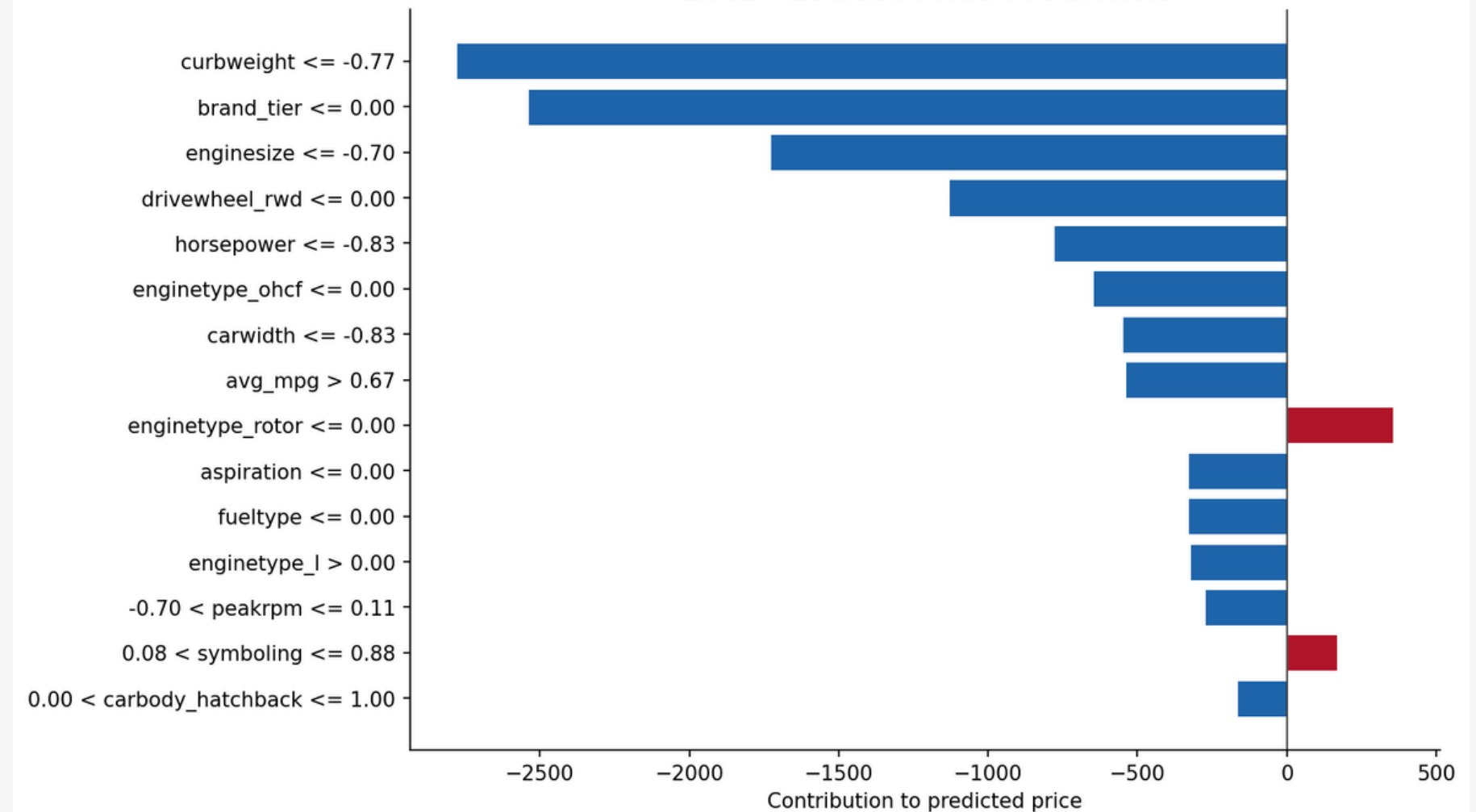


Verifying Feature Importance with LIME

LIME - Highest Price Prediction



LIME - Lowest Price Prediction



Findings & Reflection

Takeaways:

- Random Forest won across every phase (manual, PyCaret raw, PyCaret preprocessed)
- Engineered features - brand_tier, hp_per_weight, avg_mpg - were the strongest predictors

Limitations:

- Small dataset (205 rows) limits generalisation
- No manufacture year or inflation adjustment - prices shift with time and model age

Next Steps:

- Test on a larger, more varied dataset (multiple markets or time periods)
- Add contextual features (year, mileage) and tune Random Forest further



Thank you for watching!

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